MSG No. 81.37 $P_C\bar{4}$ [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[5,6,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,5,6]`

Table 4: Wyckoff site: `2d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[3,4,5,6]`

Table 5: Wyckoff site: `4e`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$[0,\ 0,\ -z]$	`[3,4]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[7,8]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ 0,\ z\right]$	`[5,6]`
4	$\left[0,\ \frac{1}{2},\ -z\right]$	`[7,8]`

Table 7: Wyckoff site: `4g`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,6]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[2,5]`
3	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[3,8]`
4	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[4,7]`

Table 8: Wyckoff site: `8h`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[2]`
3	$[y, -x, -z]$	`[3]`
4	$[-y, x, -z]$	`[4]`
5	$\left[x + \frac{1}{2}, y + \frac{1}{2}, z\right]$	`[5]`
6	$\left[\frac{1}{2} - x, \frac{1}{2} - y, z\right]$	`[6]`
7	$\left[y + \frac{1}{2}, \frac{1}{2} - x, -z\right]$	`[7]`
8	$\left[\frac{1}{2} - y, x + \frac{1}{2}, -z\right]$	`[8]`